

Global Economic Shifts

ECON-220

Trade and Resources: The Heckscher–Ohlin Model

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Questions to Consider

1. Why does the United States export agricultural products and airplanes?
2. What country has the most capital (i.e., factories) as compared with its GDP?
3. How does trade affect the earnings of labor and capital?

Introduction (part 1)

- In this lecture, we outline the Heckscher–Ohlin (HO) model, a model that assumes that trade occurs because countries have different resources.
- Canada has a large amount of land and therefore exports agricultural and forestry products, as well as petroleum.
- The United States, Western Europe, and Japan have many highly skilled workers and much capital, and these countries export sophisticated services and manufactured goods.
- Asian countries have a large number of workers and moderate but growing amounts of capital, and they export less sophisticated manufactured goods.

Introduction (part 2)

- Our first goal is to describe the Heckscher–Ohlin (HO) model of trade.
 - The specific-factors model that we studied in the previous lecture was a short-run model because *capital and land could not move between the two industries*.
 - In contrast, the HO model is a long-run model because *all factors of production can move between the industries*.

Introduction (part 3)

- Our second goal is to examine the empirical evidence on the Heckscher–Ohlin model. To obtain better predictions, we extend the model:
 - By allowing for more than two factors of production
 - By also allowing countries to differ in their technologies, as in the Ricardian model
- Both of these extensions make the predictions from the Heckscher–Ohlin model match more closely the trade patterns in the world economy today.
- The third goal of the lecture is to investigate how the opening of trade between two countries affects the payments to labor and to capital in each of them.

1) Heckscher–Ohlin Model (part 1)

- Assumptions of the Heckscher–Ohlin Model

Assumption 1: Both factors, labor and capital, can move freely between the industries.

Assumption 2: Shoe production is labor-intensive; that is, it requires more labor per unit of capital to produce shoes than computers, so $L_S/K_S > L_C/K_C$.

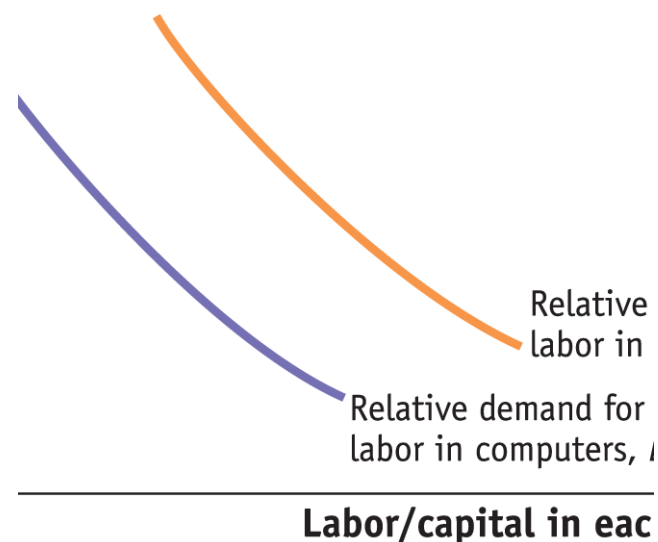
Labor Intensity of Each Industry

Shoe production being more labor-intensive than computers implies:

$$L_S/K_S > L_C/K_C$$

These two curves slope down just like regular demand curves, but in this case, they are *relative* demand curves for labor.

FIGURE 4-1



1) Heckscher–Ohlin Model (part 2)

- Assumptions of the Heckscher–Ohlin Model
- Assumption 3: Foreign is labor-abundant, by which we mean that the labor–capital ratio in Foreign exceeds that in Home,

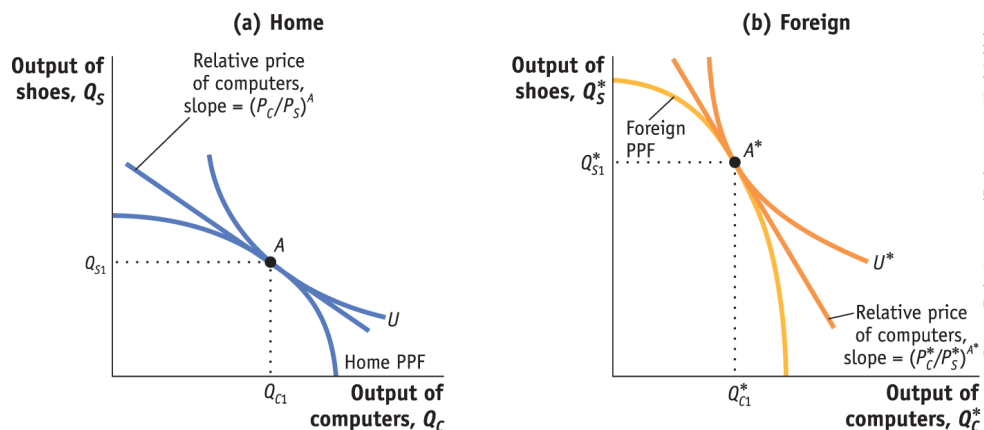
$$\bullet \quad \bar{L}^*/\bar{K}^* > \bar{L}/\bar{K}$$

- Equivalently, Home is capital-abundant, so that $\bar{K}/\bar{L} > \bar{K}^*/\bar{L}^*$.
- Assumption 4: The final outputs, shoes and computers, can be traded freely (i.e., without any restrictions) between nations, but labor and capital do not move between countries.
- Assumption 5: The technologies used to produce the two goods are identical across the countries.
- Assumption 6: Consumer tastes are the same across countries, and preferences for computers and shoes do not vary with a country's level of income.

1) Heckscher–Ohlin Model (part 3)

- No-Trade Equilibrium
- Production Possibilities Frontiers, Indifference Curves, and No-Trade Equilibrium Price

FIGURE 4-2 (1 of 3) No-Trade Equilibria in Home and Foreign



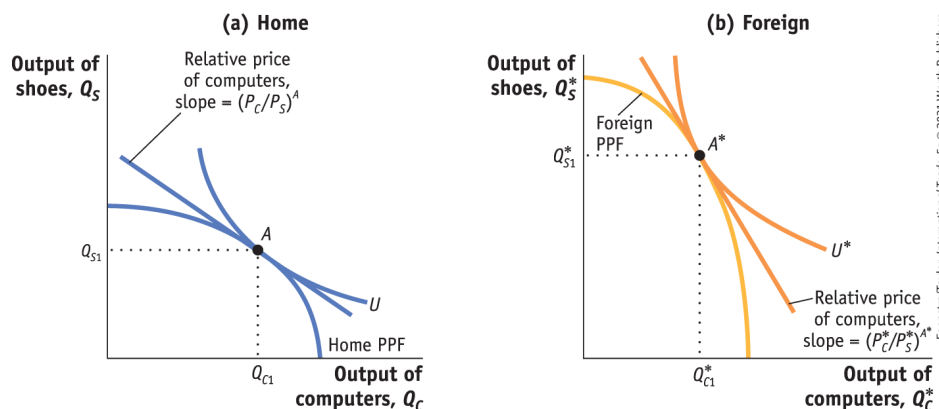
The Home production possibilities frontier (PPF) is shown in panel (a), and the Foreign PPF is shown in panel (b).

Because Home is capital-abundant and computers are capital-intensive, the Home PPF is skewed toward computers.

1) Heckscher–Ohlin Model (part 4)

- No-Trade Equilibrium
- Production Possibilities Frontiers, Indifference Curves, and No-Trade Equilibrium Price

FIGURE 4-2 (2 of 3) No-Trade Equilibria in Home and Foreign



Home preferences are summarized by the indifference curve, U .

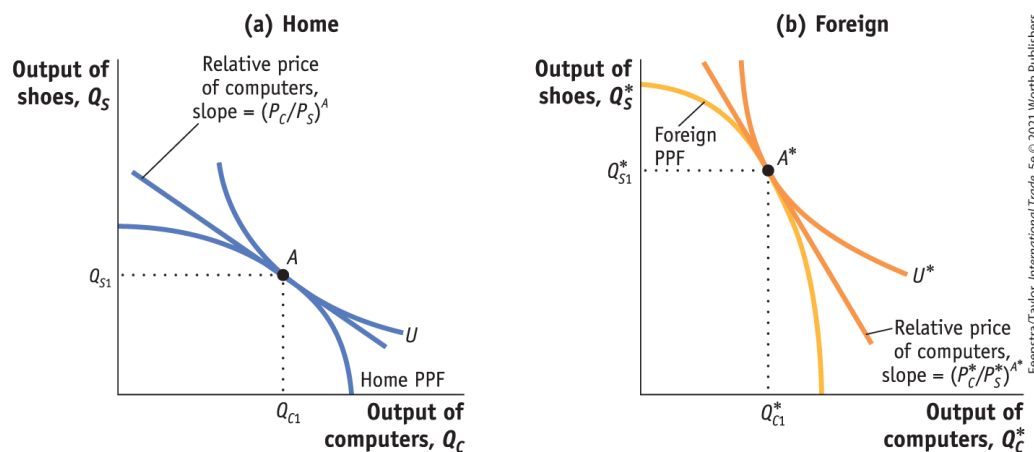
The Home no-trade (or autarky) equilibrium is at point A .

The flat slope indicates a low relative price of computers, $(P_C/P_S)^A$.

1) Heckscher–Ohlin Model (part 5)

- No-Trade Equilibrium
- Production Possibilities Frontiers, Indifference Curves, and No-Trade Equilibrium Price

FIGURE 4-2 (3 of 3) No-Trade Equilibria in Home and Foreign



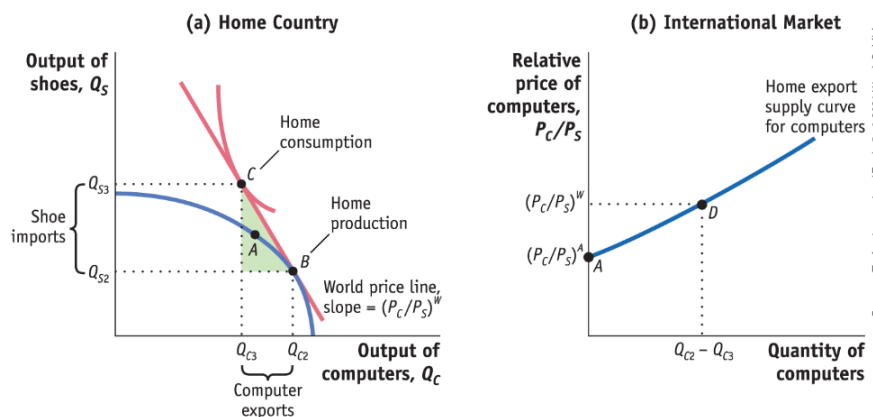
Foreign is labor-abundant and shoes are labor-intensive, so the Foreign PPF is skewed toward shoes. Foreign preferences are summarized by the indifference curve, U^* .

The Foreign no-trade equilibrium is at point A^* , with a higher relative price of computers, as indicated by the steeper slope of $(P_C^*/P_S^*)^{A^*}$.

1) Heckscher–Ohlin Model (part 6)

- No-Trade Equilibrium
- Home Equilibrium with Free Trade

FIGURE 4-3 (1 of 2) International Free-Trade Equilibrium in Home



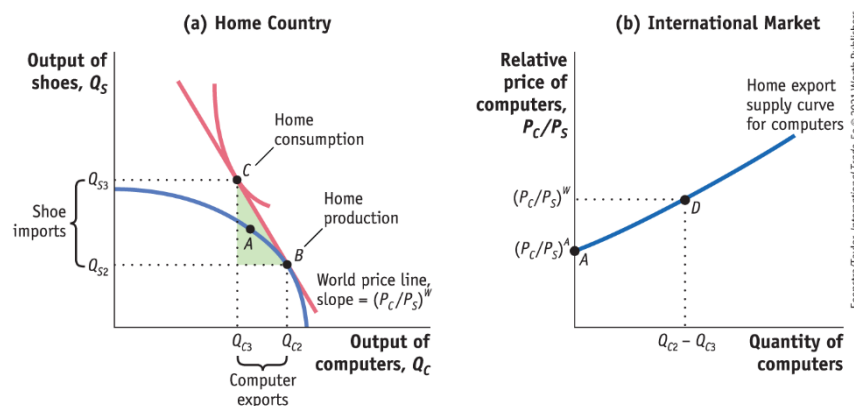
At the free-trade world relative price of computers, $(P_C/P_S)^W$, Home produces at point B in panel (a) and consumes at point C , exporting computers and importing shoes. Point A is the no-trade equilibrium.

The “trade triangle” has a base equal to the Home exports of computers (the difference between the amount produced and the amount consumed with trade, $Q_{C2} - Q_{C3}$).

1) Heckscher–Ohlin Model (part 7)

- No-Trade Equilibrium
- Home Equilibrium with Free Trade

FIGURE 4-3 (2 of 2) International Free-Trade Equilibrium in Home



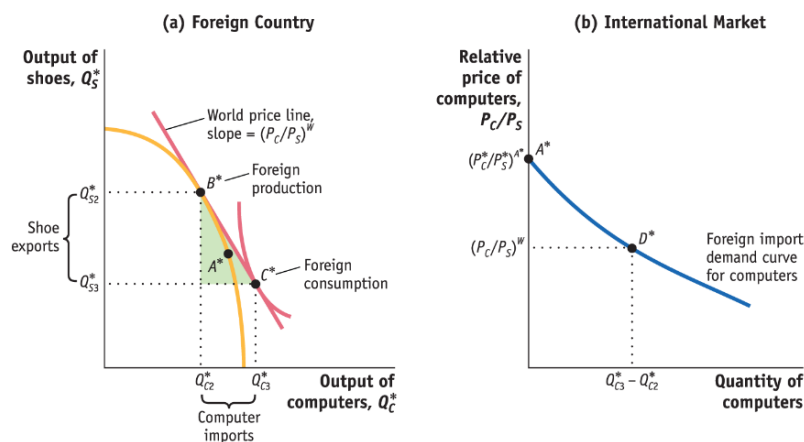
The height of this triangle is the Home imports of shoes (the difference between the amount consumed of shoes and the amount produced with trade, $Q_{S3} - Q_{S2}$).

In panel (b), we show Home exports of computers equal to zero at the no-trade relative price, $(P_C/P_S)^A$, and equal to $(Q_{C2} - Q_{C3})$ at the free-trade relative price, $(P_C/P_S)^W$.

1) Heckscher–Ohlin Model (part 8)

- No-Trade Equilibrium
- Foreign Equilibrium with Free Trade

FIGURE 4-4 (1 of 2) International Free-Trade Equilibrium in Foreign



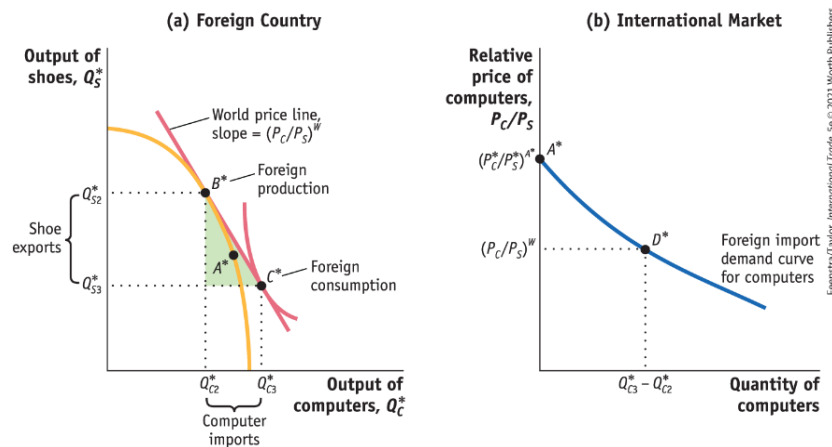
At the free-trade world relative price of computers, $(P_C/P_S)^W$, Foreign produces at point B^* in panel (a) and consumes at point C^* , importing computers and exporting shoes. Point A^* is the no-trade equilibrium.

The “trade triangle” has a base equal to Foreign imports of computers (the difference between the consumption of computers and the amount produced with trade, $Q_{C3}^* - Q_{C2}^*$).

1) Heckscher–Ohlin Model (part 9)

- No-Trade Equilibrium
- Foreign Equilibrium with Free Trade

FIGURE 4-4 (2 of 2) International Free-Trade Equilibrium in Foreign



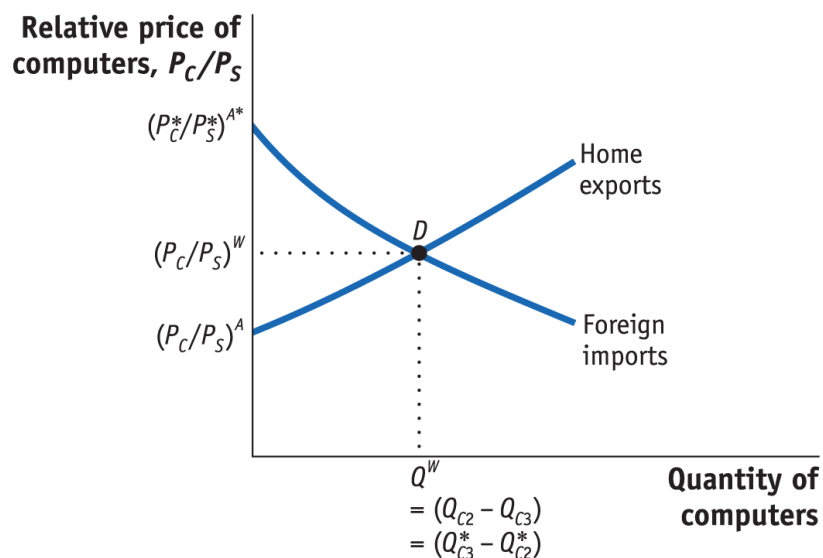
The height of this triangle is Foreign exports of shoes (the difference between the production of shoes and the amount consumed with trade, $Q_{s2}^* - Q_{s3}^*$).

In panel (b), we show Foreign imports of computers equal to zero at the no-trade relative price, $(P_c^*/P_s^*)^{A^*}$, and equal to $(Q_{c3}^* - Q_{c2}^*)$ at the free-trade relative price, $(P_c/P_s)^w$.

1) Heckscher–Ohlin Model (part 10)

- Free Trade Equilibrium
- Equilibrium Price with Free Trade

FIGURE 4-5 Determination of the Free-Trade World Equilibrium Price



The world relative price of computers in the free-trade equilibrium is determined at the intersection of the Home export supply and Foreign import demand, at point D .

At this relative price, the quantity of computers that Home wants to export, $(Q_{C2} - Q_{C3})$, just equals the quantity of computers that Foreign wants to import, $(Q_{C3}^* - Q_{C2}^*)$.

1) Heckscher–Ohlin Model (part 11)

- Free Trade Equilibrium
- Pattern of Trade
 - Home exports computers, the good that uses intensively the factor of production (capital) found in abundance at Home.
 - Foreign exports shoes, the good that uses intensively the factor of production (labor) found in abundance there.
 - This important result is called the **Heckscher–Ohlin theorem**.

1) Heckscher–Ohlin Model (part 12)

- Heckscher–Ohlin Theorem

- **Assumption 1:** Labor and capital flow freely between the industries.
- **Assumption 2:** The production of shoes is labor-intensive as compared with computer production, which is capital-intensive.
- **Assumption 3:** The amounts of labor and capital found in the two countries differ, with labor abundant in Foreign and capital abundant in Home.
- **Assumption 4:** There is free international trade in goods.
- **Assumption 5:** The technologies for producing shoes and computers are the same across countries.
- **Assumption 6:** Tastes are the same across countries.

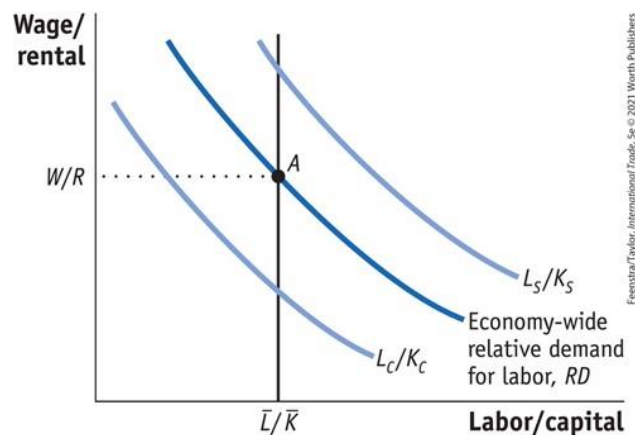
1) Heckscher–Ohlin Model (part 13)

- Heckscher–Ohlin Theorem
- Conclusions from Assumptions:
 - When Home opens to trade, its relative price of computers rises from the no-trade equilibrium relative price $(P_C/P_S)^A$ to the free-trade equilibrium price $(P_C/P_S)^W$, giving Home firms an incentive to export computers.
 - Similarly, when Foreign opens to trade, its relative price of computers falls from the no-trade equilibrium price $(P_C^*/P_S^*)^{A*}$ to the trade equilibrium price $(P_C/P_S)^W$, encouraging Foreign consumers to import computers from Home.

2) Effects of Trade on Factor Prices (part 1)

Effect of Trade on the Wage and Rental of Home
Economy-Wide Relative Demand for Labor

FIGURE 4-11 Determination of Home Wage/Rental



$$\underbrace{\frac{\bar{L}}{\bar{K}}}_{\text{Relative supply}} = \frac{L_C + L_S}{\bar{K}} = \underbrace{\frac{L_C}{K_C} \cdot \left(\frac{K_C}{\bar{K}} \right) + \frac{L_S}{K_S} \cdot \left(\frac{K_S}{\bar{K}} \right)}_{\text{Relative demand}}$$

The economy-wide relative demand for labor, RD , is an average of the L_C/K_C and L_S/K_S curves and lies between these curves.

The relative supply, L/K , is shown by a vertical line because the total amount of resources in Home is fixed.

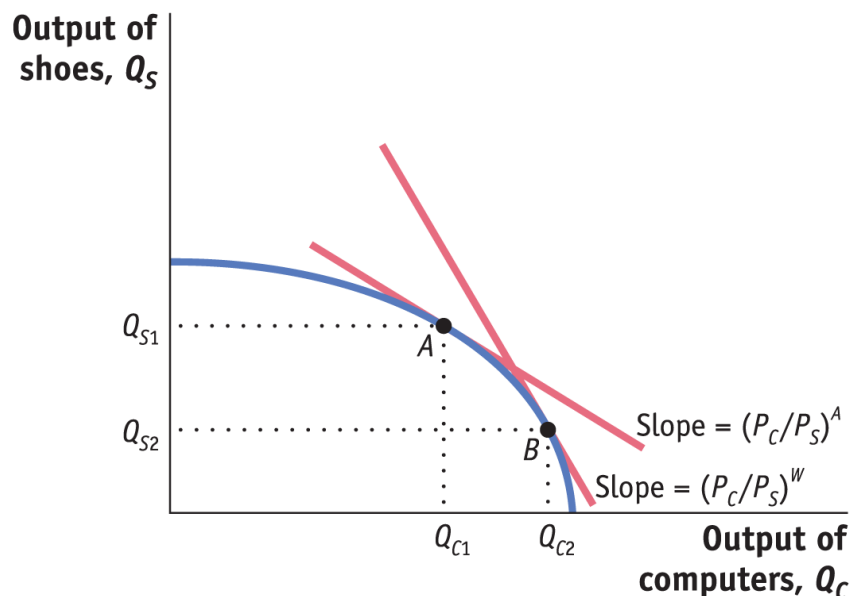
The equilibrium point A, at which relative demand RD intersects relative supply L/K , determines the wage relative to the rental, W/R .

2) Effects of Trade on Factor Prices (part 2)

Effect of Trade on the Wage and Rental of Home

Increase in the Relative Price of Computers

FIGURE 4-12 Increase in the Price of Computers



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Initially, Home is at a no-trade equilibrium at point A with a relative price of computers of $(P_C/P_S)^A$.

An increase in the relative price of computers to the world price, as illustrated by the steeper world price line, $(P_C/P_S)^W$, shifts production from point A to B.

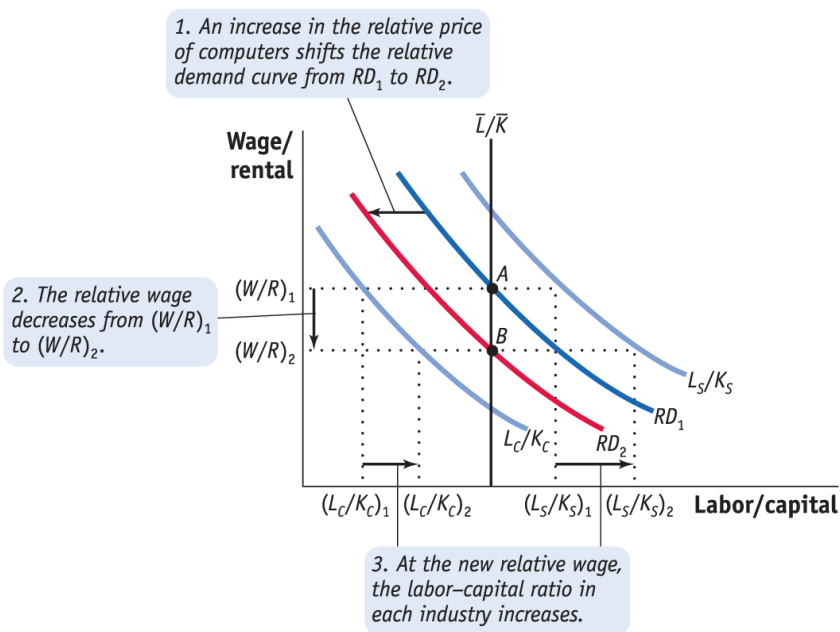
At point B, there is a higher output of computers and a lower output of shoes, $Q_{C2} > Q_{C1}$ and $Q_{S2} < Q_{S1}$.

2) Effects of Trade on Factor Prices (part 3)

Effect of Trade on the Wage and Rental of Home

Increase in the Relative Price of Computers

FIGURE 4-13 (1 of 2) Effect of a Higher Relative Price of Computers on Wage/Rental



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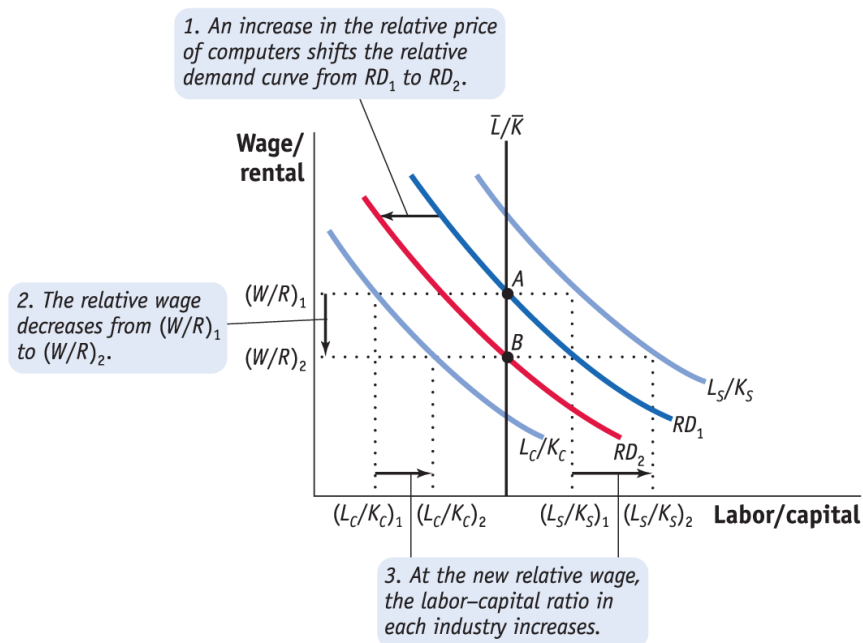
An increase in the relative price of computers shifts the economy-wide relative demand for labor, RD_1 , toward the relative demand for labor in the computer industry, L_C/K_C . The new relative demand curve, RD_2 , intersects the relative supply curve for labor at a lower relative wage, $(W/R)_2$.

2) Effects of Trade on Factor Prices (part 4)

Effect of Trade on the Wage and Rental of Home

Increase in the Relative Price of Computers

FIGURE 4-13 (2 of 2) Effect of a Higher Relative Price of Computers on Wage/Rental



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As a result, the wage relative to the rental falls from $(W/R)_1$ to $(W/R)_2$.

The lower relative wage causes both industries to increase their labor-capital ratios, as illustrated by the increase in both L_C/K_C and L_S/K_S at the new relative wage.

$$\underbrace{\frac{\bar{L}}{\bar{K}}}_{\substack{\text{Relative supply} \\ \text{No change}}} = \underbrace{\frac{L_C}{K_C} \cdot \left(\frac{K_C}{K}\right) + \frac{L_S}{K_S} \cdot \left(\frac{K_S}{K}\right)}_{\substack{\text{Relative demand} \\ \text{No change in total}}}$$

$\uparrow$ $\uparrow$ $\uparrow$ $\downarrow$

2) Effects of Trade on Factor Prices (part 5)

Determination of the Real Wage and Real Rental

Change in the Real Rental

$$R = P_C \cdot MPK_C \text{ and } R = P_S \cdot MPK_S$$

$$MPK_C = R/P_C \uparrow \text{ and } MPK_S = R/P_S \uparrow$$

Change in the Real Wage

$$W = P_C \cdot MPL_C \text{ and } W = P_S \cdot MPL_S$$

$$MPL_C = W/P_C \downarrow \text{ and } MPL_S = W/P_S \downarrow$$

2) Effects of Trade on Factor Prices (part 6)

Determination of the Real Wage and Real Rental

Stolper–Samuelson Theorem

- In the long run, when all factors are mobile, an increase in the relative price of a good will increase the real earnings of the factor used intensively in the production of that good and decrease the real earnings of the other factor.
- For our example, the **Stolper–Samuelson theorem** predicts that when Home opens to trade and faces a higher relative price of computers, the real rental on capital in Home rises and the real wage in Home falls. In Foreign, the changes in real factor prices are just the reverse.

2) Effects of Trade on Factor Prices (part 7)

Changes in the Real Wage and Rental: A Numerical Example

To illustrate the Stolper–Samuelson theorem, we use a numerical example to show how much the real wage and rental can change in response to a change in price.

Computers:

$$\text{Sales revenue} = P_c \cdot Q_c = 100$$

$$\text{Earnings of labor} = W \cdot L_c = 50$$

$$\text{Earnings of capital} = R \cdot K_c = 50$$

Shoes:

$$\text{Sales revenue} = P_s \cdot Q_s = 100$$

$$\text{Earnings of labor} = W \cdot L_s = 60$$

$$\text{Earnings of capital} = R \cdot K_s = 40$$

2) Effects of Trade on Factor Prices (part 8)

Changes in the Real Wage and Rental: A Numerical Example

Notice that shoes are more labor-intensive than computers:

- The share of total revenue paid to labor in shoes is $60/100 = 60\%$.
- This is more than that share in computers: $50/100 = 50\%$.

When Home and Foreign undertake trade, the relative price of computers rises in Home. For simplicity:

Computers: Percentage increase in price = $\Delta P_C / P_C = 10\%$

Shoes: Percentage increase in price = $\Delta P_S / P_S = 0\%$

2) Effects of Trade on Factor Prices (part 9)

Changes in the Real Wage and Rental: A Numerical Example

- The rental on capital can be calculated by taking total sales revenue in each industry, subtracting the payments to labor, and dividing by the amount of capital.
- This calculation gives us the following formulas for the rental in each industry:

$$R = \frac{P_C \cdot Q_C - W \cdot L_C}{K_C}, \text{ for computers}$$

$$R = \frac{P_S \cdot Q_S - W \cdot L_S}{K_S}, \text{ for shoes}$$

2) Effects of Trade on Factor Prices (part 10)

Changes in the Real Wage and Rental: A Numerical Example

- The price of computers has risen, so $\Delta P_C > 0$, holding fixed the price of shoes, $\Delta P_S = 0$.
- We can trace through how this affects the rental by changing P_C and W in the previous two equations:

$$R = \frac{P_C \cdot Q_C - \Delta W \cdot L_C}{K_C}, \text{ for computers}$$

$$R = \frac{0 \cdot Q_C - \Delta W \cdot L_S}{K_S}, \text{ for shoes}$$

2) Effects of Trade on Factor Prices (part 11)

Changes in the Real Wage and Rental: A Numerical Example

- It is convenient to work with percentage changes in the variables. We can introduce these terms into the preceding formulas by rewriting them as:

$$\frac{\Delta R}{R} = \left(\frac{\Delta P_C}{P_C} \right) \left(\frac{P_C \cdot Q_C}{R \cdot K_C} \right) - \left(\frac{\Delta W}{W} \right) \left(\frac{W \cdot L_C}{R \cdot K_C} \right), \text{ for computers}$$

$$\frac{\Delta R}{R} = - \left(\frac{\Delta W}{W} \right) \left(\frac{W \cdot L_S}{R \cdot K_S} \right), \text{ for shoes}$$

- Plug the above data for shoes and computers into these formulas:

$$\frac{\Delta R}{R} = 10\% \cdot \left(\frac{100}{50} \right) - \left(\frac{\Delta w}{w} \right) \left(\frac{50}{50} \right), \text{ for computers}$$

$$\frac{\Delta R}{R} = - \left(\frac{\Delta W}{W} \right) \left(\frac{60}{40} \right), \text{ for shoes}$$

2) Effects of Trade on Factor Prices (part 12)

Changes in the Real Wage and Rental: A Numerical Example

- Subtracting one equation from the other, we get

$$\begin{aligned}
 & \frac{\Delta R}{R} \quad , \text{ for computers} \\
 & = 10\% \cdot \left(\frac{100}{50} \right) - \left(\frac{\Delta W}{W} \right) \left(\frac{50}{50} \right) \\
 \text{Minus: } & \left(\frac{\Delta R}{R} \right) = 0 - \left(\frac{\Delta W}{W} \right) \left(\frac{40}{40} \right), \text{ for shoes} \\
 \hline
 \text{Equals: } & 0 = 10\% \cdot \left(\frac{100}{50} \right) + \left(\frac{\Delta w}{w} \right) \left(\frac{20}{40} \right)
 \end{aligned}$$

2) Effects of Trade on Factor Prices (part 13)

Changes in the Real Wage and Rental: A Numerical Example

- Simplifying the last line, we get $0 = 20\% + \left(\frac{\Delta W}{W}\right) \left(\frac{1}{2}\right)$.

$$\left(\frac{\Delta W}{W}\right) = \left(\frac{-20\%}{\frac{1}{2}}\right) = -40\%, \text{ change in wages}$$

- To find the change in the rental paid to capital ($\Delta R/R$), we can take our solution for $\Delta W/W = -40\%$ and plug it into the equation for the change in the rental in the shoes sector.

$$\left(\frac{\Delta R}{R}\right) = -\left(\frac{\Delta W}{W}\right) \left(\frac{60}{40}\right) = 40\% \cdot \left(\frac{60}{40}\right) = 60\% \text{ change in rental}$$

2) Effects of Trade on Factor Prices (part 14)

Changes in the Real Wage and Rental: A Numerical Example

General Equation for the Long-Run Change in Factor Prices

The long-run results of a change in factor prices can be summarized in the following equation:

$$\underbrace{\Delta W/W < 0}_{\text{Real wage falls}} < \underbrace{\Delta P_C/P_C < \Delta R/R}_{\text{Real rental increases}}, \text{ for an increase in } P_C$$

The relationship between the changes in product prices to changes in factor prices is called the “magnification effect” because it shows how changes in the prices of goods have a *magnified effect* on the earnings of factors.

$$\underbrace{\Delta R/R < \Delta P_C/P_C}_{\text{Real rental falls}} < \underbrace{P_C < 0 < \Delta W/W}_{\text{Real wage increases}}, \text{ for a decrease in } P_C \quad \underbrace{\Delta R/R < 0}_{\text{Real rental falls}} < \underbrace{\Delta P_S/P_S < \Delta W/W}_{\text{Real wage increases}}, \text{ for an increase in } P_S$$

Conclusions (part 1)

- The Heckscher–Ohlin model isolates the effect of different factor endowments across countries and determines the impact of these differences on trade patterns, relative prices of goods between countries, and factor returns.
- The Heckscher–Ohlin model predicts that countries export goods that use their abundant factor intensively.
- Leontief paradoxically found that the exports from the United States were relatively labor-intensive. This paradox was explained with later research showing that the United States was abundant in effective labor, as measured by productivity, in 1947.
- Adjusting factors of production for their productivity also explains the changing abundance and scarcity of arable land in the United States and of R&D scientists in China.

Conclusions (part 2)

- When firms have differentiated products and increasing returns to scale, the potential exists for gains from trade above and beyond those under perfect competition.
- By focusing on the factor intensities among goods (i.e., the relative amount of labor and capital used in production), the Heckscher–Ohlin (HO) model also explains who gains and who loses from the opening of trade.
- The HO model predicts real gains for the factor used intensively in the export good, whose relative price goes up with the opening of trade, and real losses for the other factor.
- Having just two factors, both of which are fully mobile between the industries, leads to a clear prediction about who gains and who loses from trade in the long run.